FLAGSHIP 05 · PERSONAL THERMAL MANAGEMENT SERIES

# LC Frost-PCM/G

## Flame Retardant PCM Cooling Vest · Antistatic & Flame-Retardant PCM Armo

**Antistatic & Flame-Retardant PCM Armor. Passive Cooling for Hazardous Frontlines.**

Engineered with high-energy-density Phase Change Material absorbing heat at constant ~23°C and encapsulated within an antistatic, flame-retardant outer shell. Designed for petrochemical, foundry, welding, and utility operations where spark hazards, explosive dust, and extreme heat converge.

### 01 Flame-Retardant & Antistatic Protection

Heavy-duty industrial twill fabric repels slag, sparks, and static charge accumulation.

### 02 Constant 22.08°C Thermal Regulation

Maintains continuous core cooling without cold shock during heavy manual labor in 35°C+ ambient heat.

### 03 High-Stability C17 Biomass Core

Thousands of thermal cycles without degradation; non-toxic and eco-friendly.

### 04 Night & Low-Light Visibility

Dual high-visibility silver reflective bands provide 360° safety in roadworks and dark industrial plants.

LC Frost-PCM/G · 3200K STUDIO MASTER



KEY METRIC 01
**22.08°C**
Non-Irritating Phase Change Comfort Zone

KEY METRIC 02
**225 J/g**
Latent Heat Storage Capacity

KEY METRIC 03
**360°**
High-Visibility Retro-Reflective Safety Bands

KEY METRIC 04
**0V**
Zero-Spark Intrinsically Safe Passive Cooling

## MATERIAL SCIENCE & MICRO-ARCHITECTURE · 3200K STUDIO MACRO INSPECTION



FR & ANTISTATIC INDUSTRIAL TWILL

### Antistatic & Flame-Retardant Ripstop Outer Shell

Specially treated antistatic and flame-retardant industrial weave dissipates electrostatic charges and resists welding splatter, enabling safe core cooling inside petrochemical refineries and gas plants.



360° RETRO-REFLECTIVE TRIM

### Dual 360° Retro-Reflective Safety Warning Bands

Wide silver retro-reflective bands wrap 360° around the chest and waist, ensuring immediate operator recognition in low-light refineries, tunnels, and night-shift roadworks.



MODULAR PCM BAYS · 0.96KG TOTAL

### Modular Internal PCM Bays & Elastic Waistband

Internal thermal compartments allow crews to swap freshly recharged 22.08°C C17 PCM packs in seconds while the wide magic-tape waistband locks the vest closely to the torso.

## ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **FR & Antistatic Outer Weave [Flame-Retardant Industrial Twill Shell]**
   Rugged ripstop fabric resists tearing, abrasions, and molten metal splatter.
2. **Elastic Magic-Tape Waistband [Heavy-Duty Cummerbund Fastener]**
   Wide industrial hook-and-loop cummerbund fits securely across all torso builds.
3. **Modular PCM Thermal Pockets [Internal C17 Thermal Pack Bays]**
   Dedicated internal compartments hold sealed PCM packs tight against the core.
4. **Dual Safety Warning Reflective Bands [360° Retro-Reflective Warning Strips]**
   3M-grade retro-reflective stripes across the chest and waist ensure field visibility.

**FIELD DEPLOYMENT | EX-RATED PETROCHEMICAL CRACKING UNIT | HIGH-HEAT ZONE**



### Petrochemical Refineries, Welding Bays & Night-Shift Utilities

In explosion-proof (EX) petrochemical zones and active welding shops, electronic batteries and brushed motors pose unacceptable ignition risks. The LC Frost-PCM/G pairs 0V passive 22.08°C phase-change cooling with an antistatic, flame-retardant outer shell and 360° reflective bands - delivering uncompromising thermal relief where electronics are forbidden.

| 0V Intrinsically Safe | 22.08°C (225 J/g) | 0.96kg |
|---|---|---|
| Ignition & Spark Hazard | Phase Change Comfort | Complete Vest Weight |

## COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-PCM/G (100% UNABBREVIATED)

| ENGINEERING GROUP | PARAMETER | VERIFIED FACTORY SPECIFICATION |
|---|---|---|
| IDENTITY & COMPLIANCE | **Model** | LC Frost-PCM/G |
| IDENTITY & COMPLIANCE | **Color** | Industrial Grey with High-Visibility Reflective Trim (customizable) |
| IDENTITY & COMPLIANCE | **Target Users** | Petrochemical, metallurgy, welding, power grid, and high-temperature night-shift workers |
| PERFORMANCE & MECHANICS | **Heat of Fusion** | 225 J/g |
| PERFORMANCE & MECHANICS | **Phase Change Temperature** | 22.08°C |
| PERFORMANCE & MECHANICS | **Cooling Duration** | 1-2 hours (reference based on labor intensity and ambient temp) |
| PERFORMANCE & MECHANICS | **Cold Storage Recharge Time** | Frozen at -18°C for 1 hour, OR refrigerated at 0-4°C for 1-2 hours |
| MATERIALS & ERGONOMICS | **Fabric Material** | Antistatic and flame-retardant fabric |
| DIMENSIONS & WARRANTY | **Sizes** | Standard Size / Plus Size |
| DIMENSIONS & WARRANTY | **Net Weight** | 0.96kg |
| DIMENSIONS & WARRANTY | **Package Dimensions** | 40 x 30 x 10cm |
| DIMENSIONS & WARRANTY | **Warranty** | 1-Year Official Industrial Warranty |

## DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-PCM/G

**Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)**

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware